lasso_psm_analysis

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1. データ読み込み

```
# ==============================================================================
# 0. 初期設定・パッケージ読み込み
# ==============================================================================
library(tinytex)
library(here)
```

```
here() starts at /Users/snakada/Documents/univ/Peru_work/code_endes
```

```
library(haven)
library(glmnet)
```

```
Warning: package 'glmnet' was built under R version 4.5.1

Loading required package: Matrix

Loaded glmnet 4.1-10
```

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.0     ✔ stringr   1.5.1
✔ ggplot2   3.5.2     ✔ tibble    3.2.1
✔ lubridate 1.9.4     ✔ tidyr     1.3.1
✔ purrr     1.0.4     
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ tidyr::expand() masks Matrix::expand()
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
✖ tidyr::pack()   masks Matrix::pack()
✖ tidyr::unpack() masks Matrix::unpack()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflict
```

```
# library(fastDummies)
library(nnet)      # 多項ロジスティック回帰
library(survey)    # 重み付き統計解析
```

```
Loading required package: grid
Loading required package: survival

Attaching package: 'survey'

The following object is masked from 'package:graphics':

    dotchart
```

```
library(did)       # Staggered DID用（必要に応じて）
library(sjlabelled)  # ラベル付きデータ処理用
```

```

Attaching package: 'sjlabelled'

The following object is masked from 'package:forcats':

    as_factor

The following object is masked from 'package:dplyr':

    as_label

The following object is masked from 'package:ggplot2':

    as_label

The following objects are masked from 'package:haven':

    as_factor, read_sas, read_spss, read_stata, write_sas, zap_labels
```

```
library(tableone)
library(fixest)   # 回帰分析の推定結果から係数表（推定値、標準誤差、t値、p値など）を抽出

# オブジェクトクリア
rm(list = ls(all = TRUE))

# デバッグモードフラグ
# debug_mode = TRUE  # デバッグモード
debug_mode = FALSE # 通常モード

# 自作関数読み込み
source("../myTools.R")
```

```

Attaching package: 'psych'

The following object is masked from 'package:did':

    sim

The following objects are masked from 'package:ggplot2':

    %+%, alpha
```

```
# ==============================================================================
# 1. データ読み込み
# ==============================================================================
# DHSデータのルートフォルダを指定
gdrive_dir <- "/Users/snakada/Library/CloudStorage/GoogleDrive-snakada@g.ecc.u-toky

# 結果保存先
save_path <- file.path(gdrive_dir, "output", "lasso_stunting2")

df_org <- readRDS(
    file.path(gdrive_dir, "output","all_data_merged.rds")
  ) %>% filter(!is.na(nt_ch_stunt))

# データベース一覧
endes_list <- readRDS(file.path(gdrive_dir, "output","endes_data_list.rds"))

# 変数ラベル一覧
endes_var_labels <- readRDS(file.path(gdrive_dir, "output", "endes_var_labels.rds"


# 介入群グループ化
df_org <- df_org %>%
  mutate(
    treatment_group = case_when(
      treatment_start == "2009" ~ 1,  # 早期介入
      treatment_start == "2011" ~ 2,  # 後期介入
      TRUE ~ 0                        # コントロール
    ),
    treatment_group = as.factor(treatment_group)
  )
# 変数ラベル取得
var_labels <- df_org %>% get_label %>% as_tibble() %>% rename(labels = value)
var_labels$varname <- names(df_org)
```

2. LASSO用データ前処理

```
# ==============================================================================
# 2. LASSO用データ前処理（介入前データのみ）
# ==============================================================================
library(gt)

# 除外変数定義(初期値)
drop_cols_init <- c(
"b3","b4","b5","b9","bidx","caseid","hc27","hhid","m1","m13","m14","m15","m1d",
"m2a","m2b","m2c","m2d","m2e","m2f","m2g","m2h","m2i","m2j","m2k","m2l","m2m",
"m3a","m3b","m3c","m3d","m3e","m3f","m3g","m3h","m3i","m3j","m3k","m3l","m3m",
"m3n","m42c","m42d","m42e","m43","m45","m60","m61","v001","v002","v003","v005",
"v008","v012","v013","v021","v022","v024","v025","v026","v106","v190","v208",
"v467b","v467c","v467d","v467f",
"wt","state","age","treated_status","treatment_start","year_treated",
"ch_report_bw","ch_report_bw","ch_size_birth","dm_cooking_fuel_traditional","dm_coo
"rh_anc_bldpres","rh_anc_median","rh_anc_bldsamp","rh_anc_median","rh_anc_numvs","
)

# 目的変数と同値となる除去変数("nt_ch_sev_stunt", "nt_ch_haz", "nt_ch_stunt"のうち2つ)
y_var_all <- c("nt_ch_stunt", "nt_ch_sev_stunt", "nt_ch_haz", "nt_ch_sev_wast", "nt
y_var <- "nt_ch_stunt"
y_var_drop <- setdiff(y_var_all, y_var)

# 除外変数定義(最終値)
drop_cols <- c(drop_cols_init, y_var_all)

# ID・管理用変数
grouping_cols <- c("caseid", "state","treated_status", "treatment_start","group_tre
                   "wt", "treatment_group", y_var)

# 1. LASSO用のデータセット定義
df_lasso <- df_org

cat("介入前データの群別サンプル数:\n")
```

```
介入前データの群別サンプル数:
```

```
print(table(df_lasso$treatment_group))
```

```

    0     1     2 
24737  3543  3792 
```

```
# 4. NAの多い列を除去（10%以上NA）
initial_cols <- ncol(df_lasso)
keep_cols <- names(df_lasso)[sapply(df_lasso, function(x) sum(is.na(x)) <= nrow(df_
df_lasso <- df_lasso[, keep_cols]
cat("NA値の多い列を", initial_cols - ncol(df_lasso), "個削除しました\n")
```

```
NA値の多い列を 139 個削除しました
```

```
# 5. 共変量名の再定義（NA除去後）
all_drop_cols <- intersect(drop_cols, names(df_lasso))
grouping_cols <- intersect(grouping_cols, names(df_lasso))
covariate_names <- setdiff(names(df_lasso), c(all_drop_cols, grouping_cols))


# 6. 共変量のNAがある行を除去
# df_lasso <- df_lasso[, covariate_names]
complete_idx <- complete.cases(df_lasso)
initial_rows <- nrow(df_lasso)
df_lasso <- df_lasso[complete_idx, ]
cat("NA値を含む行を", initial_rows - nrow(df_lasso), "行削除しました\n")
```

```
NA値を含む行を 7958 行削除しました
```

```
cat("最終的な介入前データサイズ:", nrow(df_lasso), "行 ×", ncol(df_lasso), "列\n\n")
```

```
最終的な介入前データサイズ: 24114 行 × 182 列
```

```
# 共変量のラベル保存
covariate_names_label <- var_labels %>%
  filter(varname %in% covariate_names)
# print文の代替
covariate_names_label %>%
  gt() %>%
  cols_width(
    1 ~ pct(80),
    2 ~ pct(20)
  )
```

labels	varname
Skilled assistance during ANC	rh_anc_pvskill
Attended 4+ ANC visits	rh_anc_4vs
Number of months pregnant at time of first ANC visit	rh_anc_moprg
Attended ANC <4 months of pregnancy	rh_anc_4mo
Took iron tablet/syrup during pregnancy of last birth	rh_anc_iron
Took intestinal parasite drugs during pregnancy of last birth	rh_anc_parast
Problem health care access: permission to go	rh_prob_permit
Problem health care access: getting money	rh_prob_money
Problem health care access: distance to facility	rh_prob_dist
Problem health care access: not wanting to go alone	rh_prob_alone
At least one problem in accessing health care	rh_prob_minone
	year
Attended or completed at least secondary education	rc_edu_acceptable
Literate - higher than secondary or can read part or whole sentence	rc_litr
Accesses none of the three media at least once a week	rc_media_none
Occupation among those employed in the past 12 months	rc_occup
Work in agriculture in the past 12 months	rc_agri
Have any health insurance	rc_hins_any
Smokes any type of tobacco	rc_tobc_smk_any
ARI symptoms in the 2 weeks before the survey	ch_ari
Fever symptoms in the 2 weeks before the survey	ch_fever
Started breastfeeding within one hour of birth - last-born in the past 2 years	nt_bf_start_1hr
Started breastfeeding within one day of birth - last-born in the past 2 years	nt_bf_start_1day
Received a prelacteal feed - last-born in the past 2 years ever breast fed (without birth record data)	nt_bf_prelac_nobirthrecord
Drank from a bottle with a nipple yesterday - under 2 years	nt_bottle
Age-appropriately breastfed - last-born under 2 years	nt_ageapp_bf
Child given infant formula in day/night before survey - last-born under 2 years	nt_formula
Child given other liquids in day/night before survey- last-born under 2 years	nt_liquids
Child given fortified baby food in day/night before survey- last-born under 2 years	nt_bbyfood
Child given grains in day/night before survey- last-born under 2 years	nt_grains
Child given vitamin A rich food in day/night before survey- last-born under 2 years	nt_vita
Child given other fruits or vegetables in day/night before survey- last-born under 2 years	nt_frtveg
Child given roots or tubers in day/night before survey- last-born under 2 years	nt_root
Child given legumes or nuts in day/night before survey- last-born under 2 years	nt_nuts
Child given meat, fish, shellfish, or poultry in day/night before survey- last-born under 2 years	nt_meatfish
Child given eggs in day/night before survey- last-born under 2 years	nt_eggs
Child given cheese, yogurt, or other milk products in day/night before survey- last-born under 2 years	nt_dairy
Child given any solid or semisolid food in day/night before survey- last-born under 2 years	nt_solids
Child given milk or milk products- last-born 6-23 months	nt_fed_milk
Child with minimum dietary diversity, 5 out of 8 food groups- last-born 6-23 months	nt_mdd
Child with minimum meal frequency- last-born 6-23 months	nt_mmf
Child with minimum acceptable diet- last-born 6-23 months	nt_mad
Youngest children age 6-23 mos living with mother given Vit A rich food	nt_ch_micro_vaf
Current marital status	ms_mar_stat
First marriage by age 15	ms_afm_15
First marriage by age 18	ms_afm_18
First marriage by age 20	ms_afm_20
First marriage by age 22	ms_afm_22
First marriage by age 25	ms_afm_25
	ms_age
Never had sex	ms_sex_never
First sex by age 15	ms_afs_15
First sex by age 18	ms_afs_18
First sex by age 20	ms_afs_20
First sex by age 22	ms_afs_22
First sex by age 25	ms_afs_25
Child's stool was disposed of appropriately among youngest children under age 2 living with mother	ch_stool_safe
Number of children under 12 years old	dm_children_under12
type of cooking fuel	dm_cooking_fuel
Agree that husband is justified in hitting or beating his wife if she burns food	dm_dvjustify_burn
Agree that husband is justified in hitting or beating his wife if she argues with him	dm_dvjustify_argue
Agree that husband is justified in hitting or beating his wife if she goes out without telling him	dm_dvjustify_goout
Agree that husband is justified in hitting or beating his wife if she neglects the children	dm_dvjustify_neglect
Agree that husband is justified in hitting or beating his wife if she refuses to have sexual intercourse with him	dm_dvjustify_refusesex
Agree that husband is justified in hitting or beating his wife for at least one of the reasons	dm_dvjustify_onereas
Believe a woman is justified to refuse sex with her husband if she knows he's having sex with other women	dm_justify_refusesex
living in small town	dm_town
living in medium city	dm_city
living in capital city	dm_capital
poorest wealth quintile	dm_poorest
poorer wealth quintile	dm_poorer
middle wealth quintile	dm_middle
richer wealth quintile	dm_richer
richest wealth quintile	dm_richest
agricultural land (hectare)	dm_ag_land_ha
owns cattle	dm_cattle_own
owns goats	dm_goat_own
owns sheep	dm_sheep_own
owns poultry	dm_poultry_own
owns refrigerator	dm_refrigerator
female headed household	dm_female_headed
Age of head of household	dm_age_of_hh_head
Mild anemia - child 6-59 months	nt_ch_mild_anem

```
# ********** デバッグ用 **********
if (debug_mode) {
  cat("=== デバッグモード: 変数の詳細確認 ===\n")
  cat("共変量数:", length(covariate_names), "\n")
  cat("共変量:\n")

  # ダミーでない変数を抽出
  non_dummy <- list()
  temp <- df_lasso[, covariate_names]
  for (var in names(temp)) {
    if (!is_dummy_var(temp[[var]])) {
      non_dummy <- append(non_dummy, var)
    } else {
      cat("ダミー変数:", var)
      print(table(temp[[var]], useNA = "always"))
      cat("\n")
    }
  }
  cat("ダミー変数でない変数数:", length(non_dummy), "\n")
  cat("ダミー変数でない変数:\n")
  print(non_dummy)
  # non_dummy %>%
  #   gt() %>%
  #   cols_width(
  #     1 ~ pct(100)
  #   )
}
# ********** デバッグ用 **********
```

3. LASSOによる変数選択

```
library(gt)
# ==============================================================================
# 3. LASSO変数選択（介入前データ）
# ==============================================================================
# 完全に一から再開
cat("=== データ処理を再開 ===\n")
```

```
=== データ処理を再開 ===
```

```
cat("df_lasso の次元:", dim(df_lasso), "\n")
```

```
df_lasso の次元: 24114 182
```

```
cat("covariate_names の数:", length(covariate_names), "\n")
```

```
covariate_names の数: 83
```

```
# Step 1: model.matrix作成
cat("Step 1: model.matrix作成中...\n")
```

```
Step 1: model.matrix作成中...
```

```
X_matrix <- model.matrix(~ . - 1, data = df_lasso[, covariate_names, drop=FALSE])
Y <- df_lasso$nt_ch_stunt

cat("=== 問題列の除去 ===\n")
```

```
=== 問題列の除去 ===
```

```
# Step 1: 標準偏差が0の列を特定・除去
col_sds <- apply(X_matrix, 2, sd, na.rm = TRUE)
zero_sd_cols <- is.na(col_sds) | col_sds == 0
cat("除去する定数列の数:", sum(zero_sd_cols), "\n")
```

```
除去する定数列の数: 4
```

```
if(sum(zero_sd_cols) > 0) {
  cat("除去する列名:\n")
  print(colnames(X_matrix)[zero_sd_cols])

  # 定数列を除去
  X_matrix_clean <- X_matrix[, !zero_sd_cols, drop = FALSE]
} else {
  X_matrix_clean <- X_matrix
}
```

```
除去する列名:
[1] "nt_bbyfood"      "nt_solids"       "nt_ch_micro_vaf" "ms_sex_never"
```

```
cat("列除去後のX_matrix次元:", dim(X_matrix_clean), "\n")
```

```
列除去後のX_matrix次元: 24114 88
```

```
# Step 2: スケーリング実行
cat("=== クリーンなデータでスケーリング ===\n")
```

```
=== クリーンなデータでスケーリング ===
```

```
X_scaled <- scale(X_matrix_clean)

cat("スケーリング後の次元:", dim(X_scaled), "\n")
```

```
スケーリング後の次元: 24114 88
```

```
cat("スケーリング後のNA数:", sum(is.na(X_scaled)), "\n")
```

```
スケーリング後のNA数: 0
```

```
# Step 3: まだNAがある場合の詳細チェック
if(sum(is.na(X_scaled)) > 0) {
  cat("=== 残存するNA問題の詳細分析 ===\n")

  # 各列のNA数をチェック
  col_na_counts <- apply(X_scaled, 2, function(x) sum(is.na(x)))
  problematic_cols <- col_na_counts > 0

  cat("NA値を含む列数:", sum(problematic_cols), "\n")
  if(sum(problematic_cols) > 0) {
    cat("問題のある列:\n")
    print(col_na_counts[problematic_cols])

    # 問題のある列を除去
    X_scaled <- X_scaled[, !problematic_cols, drop = FALSE]
    cat("問題列除去後の次元:", dim(X_scaled), "\n")
  }
}

# Step 4: 最終確認
cat("=== 最終確認 ===\n")
```

```
=== 最終確認 ===
```

```
cat("X_scaled次元:", dim(X_scaled), "\n")
```

```
X_scaled次元: 24114 88
```

```
cat("X_scaledのNA数:", sum(is.na(X_scaled)), "\n")
```

```
X_scaledのNA数: 0
```

```
cat("Y長さ:", length(Y), "\n")
```

```
Y長さ: 24114
```

```
cat("Y分布:", table(Y), "\n")
```

```
Y分布: 19824 4290
```

```
# データが正常な場合のLASSO実行
if(nrow(X_scaled) > 0 && ncol(X_scaled) > 0 && sum(is.na(X_scaled)) == 0) {
  cat("=== LASSO実行 ===\n")
  set.seed(123)

  cv_model <- cv.glmnet(X_scaled, Y,
                        alpha = 1,
                        family = "multinomial",
                        nfolds = 3)

  cat("LASSO実行成功！\n")
} else {
  cat("まだデータに問題があります\n")
}
```

```
=== LASSO実行 ===
LASSO実行成功！
```

```
cat("最適λ (lambda.min):", cv_model$lambda.min, "\n")
```

```
最適λ (lambda.min): 0.001420552
```

```
cat("1se λ (lambda.1se):", cv_model$lambda.1se, "\n\n")
```

```
1se λ (lambda.1se): 0.002990126
```

```
# 選択された変数の抽出
coef_min <- coef(cv_model, s = "lambda.min")
selected_vars <- c()
for(i in 1:length(coef_min)) {
  coef_matrix <- as.matrix(coef_min[[i]])
  class_vars <- rownames(coef_matrix)[coef_matrix[,1] != 0 & rownames(coef_matrix)
  selected_vars <- union(selected_vars, class_vars)
}

cat("LASSO選択変数数:", length(selected_vars), "\n")
```

```
LASSO選択変数数: 63
```

```
if(length(selected_vars) > 0) {
  # ********** デバッグ用 **********
  if (debug_mode) {
    cat("選択された変数:\n")
    for(i in 1:length(selected_vars)) {
      cat(i, ":", selected_vars[i], "\n")
    }
  }
  # ********** デバッグ用 **********

  # 変数の重要度確認
  coef_importance <- abs(as.matrix(coef(cv_model, s = "lambda.min")[[1]]))
  top_vars <- head(sort(coef_importance[,1], decreasing = TRUE), 20)
  coef_importance <- coef_importance %>%
    as_tibble(rownames = "labels") %>%
    filter(labels %in% selected_vars) %>%
    left_join(var_labels, by = join_by("labels" == "varname"))
  saveRDS(coef_importance, file.path(save_path, "lasso_selected_vars_importance.rds

  # 列幅を指定して結果出力
  coef_importance %>%
    gt() %>%
    cols_width(
      1 ~ pct(15),
      2 ~ pct(15),
      3 ~ pct(70)
    ) %>%
    fmt_number(
      columns = c("lambda.min"),
      decimals = 4  # 小数点以下4桁
    )
} else {
  stop("変数が選択されませんでした。λの値を調整してください。")
}
```

labels	lambda.min	labels.y
rh_anc_pvskill	0.0172	Skilled assistance during ANC
rh_anc_4vs	0.0179	Attended 4+ ANC visits
rh_anc_4mo	0.0179	Attended ANC <4 months of pregnancy
rh_anc_iron	0.0025	Took iron tablet/syrup during pregnancy of last birth
rh_anc_parast	0.0082	Took intestinal parasite drugs during pregnancy of last birth
rh_prob_permit	0.0152	Problem health care access: permission to go
rh_prob_dist	0.0059	Problem health care access: distance to facility
rh_prob_minone	0.0149	At least one problem in accessing health care
year2007	0.0279	NA
year2008	0.0293	NA
year2009	0.0245	NA
year2010	0.0176	NA
year2011	0.0238	NA
year2012	0.0002	NA
year2014	0.0148	NA
year2015	0.0227	NA
year2016	0.0431	NA
rc_edu_acceptable	0.0767	Attended or completed at least secondary education
rc_litr	0.0432	Literate - higher than secondary or can read part or whole sentence
rc_media_none	0.0135	Accesses none of the three media at least once a week
rc_occup	0.0056	Occupation among those employed in the past 12 months
rc_agri	0.0292	Work in agriculture in the past 12 months
rc_hins_any	0.0128	Have any health insurance
rc_tobc_smk_any	0.0136	Smokes any type of tobacco
ch_ari	0.0026	ARI symptoms in the 2 weeks before the survey
nt_bf_start_1hr	0.0179	Started breastfeeding within one hour of birth - last-born in the past 2 years

nt_bf_start_1day	0.0684	Started breastfeeding within one day of birth - last-born in the past 2 years
nt_bf_prelac_nobirthrecord	0.0034	Received a prelacteal feed - last-born in the past 2 years ever breast fed (without birth record data)
nt_bottle	0.0265	Drank from a bottle with a nipple yesterday - under 2 years
nt_ageapp_bf	0.0010	Age-appropriately breastfed - last-born under 2 years
nt_formula	0.0074	Child given infant formula in day/night before survey - last-born under 2 years
nt_liquids	0.0069	Child given other liquids in day/night before survey- last-born under 2 years
nt_grains	0.0135	Child given grains in day/night before survey- last-born under 2 years
nt_frtveg	0.0093	Child given other fruits or vegetables in day/night before survey- last-born under 2 years
nt_root	0.0080	Child given roots or tubers in day/night before survey- last-born under 2 years
nt_nuts	0.0079	Child given legumes or nuts in day/night before survey- last-born under 2 years
nt_eggs	0.0073	Child given eggs in day/night before survey- last-born under 2 years
nt_dairy	0.0192	Child given cheese, yogurt, or other milk products in day/night before survey- last-born under 2 years
nt_fed_milk	0.0576	Child given milk or milk products- last-born 6-23 months
nt_mmf	0.0086	Child with minimum meal frequency- last-born 6-23 months
ms_mar_stat	0.0049	Current marital status
ms_afm_15	0.0103	First marriage by age 15
ms_afm_22	0.0003	First marriage by age 22
ms_afs_15	0.0031	First sex by age 15
ms_afs_18	0.0087	First sex by age 18
ms_afs_25	0.0008	First sex by age 25
ch_stool_safe	0.0055	Child's stool was disposed of appropriately among youngest children under age 2 living with mother
dm_children_under12	0.0815	Number of children under 12 years old
dm_cooking_fuel	0.0108	type of cooking fuel
dm_dvjustify_argue	0.0018	Agree that husband is justified in hitting or beating his wife if she argues with him
dm_justify_refusesex	0.0113	Believe a woman is justified to refuse sex with her husband if she knows he's having sex with other women
dm_town	0.0129	living in small town
dm_capital	0.0394	living in capital city
dm_poorest	0.0966	poorest wealth quintile
dm_poorer	0.0289	poorer wealth quintile
dm_richer	0.0576	richer wealth quintile
dm_richest	0.0930	richest wealth quintile
dm_ag_land_ha	0.0042	agricultural land (hectare)
dm_cattle_own	0.0167	owns cattle
dm_goat_own	0.0030	owns goats
dm_sheep_own	0.0499	owns sheep
dm_refrigerator	0.0582	owns refrigerator
dm_age_of_hh_head	0.0243	Age of head of household

4. 変数を上位20個に絞り込んだ上で、時間不変性を確認

```
# 1. LASSO結果から上位20変数を抽出（年ダミー除外）
selected_predictors <- coef_importance %>%
  filter(!str_detect(labels, "^year20[0-9]{2}$")) %>%
  arrange(desc(lambda.min))

df_selected_indicators <- df_lasso[ , c(selected_predictors$labels, "year")]


# 2. 各変数の年次変動を評価する関数
# ****************************************************************************
#' @title calculate_temporal_stability
#' @description 指定された変数について、年別の平均値を計算し、変動係数（CV）を用いて時間不変性を
#' @param df データフレーム
#' @param var_names 評価する変数名のベクトル
#' @param year_col 年を示す列名（デフォルトは "year"）
#' @return 各変数の平均値、標準偏差、変動係数、範囲ベースの変動指標、時間安定性の分類を含むデータ
# ****************************************************************************
calculate_temporal_stability <- function(df, var_names, year_col = "year") {

  # 年別平均値を計算
  yearly_means <- df %>%
    group_by(!!sym(year_col)) %>%
    summarise(across(all_of(var_names), ~ mean(.x, na.rm = TRUE)), .groups = "drop"

  # 各変数の変動係数（CV）と時間安定性指標を計算
  stability_results <- map_dfr(var_names, function(var) {
    yearly_values <- yearly_means[[var]]

    # 基本統計量
    mean_val <- mean(yearly_values, na.rm = TRUE)
    sd_val <- sd(yearly_values, na.rm = TRUE)
    cv_val <- sd_val / mean_val

    # 範囲ベースの変動指標
    min_val <- min(yearly_values, na.rm = TRUE)
    max_val <- max(yearly_values, na.rm = TRUE)
    range_ratio <- (max_val - min_val) / mean_val

    # 時間安定性の分類
    stability_category <- case_when(
      cv_val < 0.05 ~ "stabilit: very high",
      cv_val < 0.10 ~ "stabilit: high",
      cv_val < 0.20 ~ "stability: moderate",
      cv_val < 0.50 ~ "stability: low",
      TRUE ~ "unstable"
    )

    tibble(
      variable = var,
      mean_across_years = mean_val,
      sd_across_years = sd_val,
      cv_coefficient = cv_val,
      range_ratio = range_ratio,
      stability_category = stability_category
    )
  })

  # CV昇順でソート
  stability_results <- stability_results %>%
    arrange(cv_coefficient)

  return(stability_results)
}
# --- 関数定義ここまで ---

# 3. 関数を用いて選択された変数の時間安定性を評価
df_stability <- calculate_temporal_stability(df_lasso, selected_predictors$labels)

# 4. 結果を表示
df_stability %>%
  gt() %>%
  cols_width(
    1 ~ pct(20),
    2 ~ pct(15),
    3 ~ pct(15),
    4 ~ pct(15),
    5 ~ pct(15),
    6 ~ pct(20)
  ) %>%
  fmt_number(
    columns = c("mean_across_years", "sd_across_years", "cv_coefficient", "range_ra
    decimals = 4  # 小数点以下4桁
  )
```

variable	mean_across_years	sd_across_years	cv_coefficient	range_ratio	stabili
nt_ageapp_bf	0.9953	0.0019	0.0019	0.0068	stabili
ms_afs_25	0.9558	0.0040	0.0042	0.0122	stabili
rh_anc_4vs	0.9614	0.0090	0.0094	0.0266	stabili
nt_bf_start_1day	0.9244	0.0124	0.0134	0.0399	stabili
rc_litr	0.9460	0.0145	0.0153	0.0539	stabili
dm_age_of_hh_head	3.5648	0.0598	0.0168	0.0484	stabili
nt_mmf	0.9238	0.0161	0.0175	0.0625	stabili
nt_fed_milk	0.8498	0.0149	0.0175	0.0594	stabili
dm_justify_refusesex	0.9547	0.0168	0.0176	0.0603	stabili
ms_afs_18	0.5478	0.0099	0.0180	0.0520	stabili
ms_mar_stat	1.8549	0.0374	0.0201	0.0567	stabili
nt_nuts	0.4371	0.0101	0.0231	0.0728	stabili
nt_bf_prelac_nobirthrecord	0.7443	0.0176	0.0236	0.0946	stabili
nt_grains	0.9028	0.0221	0.0245	0.0707	stabili
nt_root	0.8109	0.0248	0.0306	0.1014	stabili
ms_afm_22	0.6662	0.0215	0.0323	0.0931	stabili
nt_eggs	0.5715	0.0218	0.0382	0.1183	stabili
dm_children_under12	2.1936	0.0928	0.0423	0.1480	stabili
nt_bf_start_1hr	0.5636	0.0244	0.0433	0.1322	stabili
nt_frtveg	0.6880	0.0309	0.0449	0.1329	stabili
rh_anc_4mo	0.7461	0.0358	0.0480	0.1546	stabili
rh_prob_minone	0.7874	0.0388	0.0493	0.1538	stabili
rh_anc_iron	0.8733	0.0479	0.0549	0.1766	stabili
nt_liquids	0.7624	0.0455	0.0597	0.1822	stabili
dm_poorer	0.2908	0.0176	0.0605	0.2047	stabili
rh_anc_pvskill	0.4154	0.0324	0.0779	0.3061	stabili
nt_bottle	0.4749	0.0407	0.0856	0.2929	stabili
ms_afs_15	0.1209	0.0113	0.0931	0.2638	stabili
rh_prob_dist	0.4350	0.0420	0.0966	0.3024	stabili
rh_anc_parast	0.0269	0.0026	0.0968	0.3795	stabili
nt_dairy	0.3921	0.0462	0.1179	0.3733	stabili moder
rh_prob_permit	0.1310	0.0159	0.1216	0.3549	stabili moder
rc_media_none	0.1522	0.0210	0.1376	0.5012	stabili moder
rc_edu_acceptable	0.5145	0.0721	0.1401	0.4596	stabili moder
dm_richer	0.1635	0.0233	0.1426	0.4953	stabili moder
dm_richest	0.0951	0.0153	0.1608	0.4218	stabili moder
dm_cooking_fuel	6.4327	1.1481	0.1785	0.5587	stabili moder
rc_occup	2.9224	0.6449	0.2207	0.7187	stabili
ms_afm_15	0.0524	0.0117	0.2232	0.6599	stabili
ch_stool_safe	0.2448	0.0639	0.2611	0.7913	stabili
dm_refrigerator	0.3573	0.0946	0.2648	0.7514	stabili
dm_town	0.2188	0.0586	0.2678	0.7084	stabili
dm_capital	0.0882	0.0246	0.2796	0.8052	stabili
rc_tobc_smk_any	0.0194	0.0058	0.2970	0.9772	stabili
dm_sheep_own	0.1163	0.0346	0.2976	0.9667	stabili
rc_agri	0.1862	0.0585	0.3142	0.9482	stabili
rc_hins_any	0.6676	0.2116	0.3169	0.9557	stabili
dm_cattle_own	0.1376	0.0447	0.3248	0.9758	stabili
dm_dvjustify_argue	0.0085	0.0029	0.3426	1.1588	stabili
ch_ari	0.0776	0.0285	0.3673	1.0809	stabili
dm_goat_own	0.0259	0.0101	0.3914	1.0552	stabili
dm_poorest	0.1944	0.0801	0.4117	1.0546	stabili
nt_formula	0.0988	0.0466	0.4711	1.3402	stabili
dm_ag_land_ha	2.7930	1.4180	0.5077	1.4174	unstal

5. 変数を上位20個に絞り込んだ上で、GLM実行

```
library(survey)

# 1. LASSO/年次安定性の結果から上位20変数を抽出（年ダミー除外）
selected_predictors <- df_stability %>%
  slice_head(n = 21) %>%
  pull(variable)

# 2. sampling weightの年次補正
# 年ごとの元 v005 合計を算出
year_totals <- df_lasso %>%
  group_by(year) %>%
  summarise(total_v005 = sum(v005, na.rm = TRUE), .groups = 'drop')

# 正規化済みサンプリングウェイトの作成
df_lasso <- df_lasso %>%
  select(-any_of("total_v005")) %>%  # 存在する場合のみ削除
  left_join(year_totals, by = "year") %>%
  mutate(
    # 元のDHSウェイト
    dhs_weight = v005 / 1e6,
    # 年次正規化係数：全期間対象の標本サイズ比（例：年の全体ウェイト比）
    year_norm = total_v005 / sum(year_totals$total_v005),
    # 正規化済みウェイト
    sampling_weight_norm = dhs_weight / year_norm,
    # 極端値トリミング
    sampling_weight_trimmed = pmax(
      pmin(sampling_weight_norm,
           quantile(sampling_weight_norm, 0.99, na.rm = TRUE)),
      quantile(sampling_weight_norm, 0.01, na.rm = TRUE)
    )
  )

# 3. surveydesignオブジェクトの作成
# Survey designオブジェクトの作成
svy_design <- svydesign(
  ids = ~v021,          # Primary Sampling Unit（クラスター）
  strata = ~v022,       # 層化変数
  weights = ~sampling_weight_trimmed,
  data = df_lasso,
  nest = TRUE           # 入れ子構造
)
# 孤立PSUの処理
options(survey.lonely.psu = "adjust")

# 4. GLMの実施（ロジスティク）
# GLMによるモデル再構築
final_formula <- as.formula(
  paste("nt_ch_stunt ~",
        paste(selected_predictors, collapse = " + "),
        "+ as.factor(year)")
)
# ********** デバッグ用 **********
if (debug_mode) {
  cat("=== デバッグモード: 最終モデルの確認 ===\n")
  cat("最終モデルの式:\n")
  print(final_formula)
}
# ********** デバッグ用 **********

# Survey-weighted GLM
final_model_svy <- svyglm(
  final_formula,
  design = svy_design,
  family = binomial
)
```

```
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
```

```
# 5. 結果出力
library(gtsummary)
```

```
Warning: package 'gtsummary' was built under R version 4.5.1
```

```
library(broom.helpers)
```

```
Warning: package 'broom.helpers' was built under R version 4.5.1


Attaching package: 'broom.helpers'

The following objects are masked from 'package:gtsummary':

    all_categorical, all_continuous, all_contrasts, all_dichotomous,
    all_interaction, all_intercepts
```

```
# gtsummaryを使用してgtテーブルを作成
final_model_svy %>%
  tbl_regression(
    exponentiate = TRUE,  # オッズ比を表示したい場合
    intercept = TRUE,     # 切片を表示したい場合
    pvalue_fun = function(x) style_pvalue(
      x, digits = 3, stars = TRUE
    )
  ) %>%
  add_significance_stars() %>%
  as_gt() %>%
  gt::tab_header(
    title    = "Survey-weighted Logistic Regression",
    subtitle = "※ *p<0.05, **p<0.01, ***p<0.001"
  )
```

Survey-weighted Logistic Regression

※ *p<0.05, **p<0.01, ***p<0.001

Characteristic	OR[1]	SE
(Intercept)	3.70**	0.451
Age-appropriately breastfed - last-born under 2 years	0.68	0.347
First sex by age 25	0.80	0.136
Attended 4+ ANC visits	0.88	0.129
Started breastfeeding within one day of birth - last-born in the past 2 years	0.71**	0.104
Literate - higher than secondary or can read part or whole sentence	0.35***	0.094
Age of head of household	0.90***	0.021
Child with minimum meal frequency- last-born 6-23 months	1.06	0.102
Child given milk or milk products- last-born 6-23 months	0.53***	0.110
Believe a woman is justified to refuse sex with her husband if she knows he's having sex with other women	0.76*	0.114
First sex by age 18	1.03	0.058
Current marital status	0.97	0.027
Child given legumes or nuts in day/night before survey- last-born under 2 years	1.26***	0.055
Received a prelacteal feed - last-born in the past 2 years ever breast fed (without birth record data)	1.32**	0.097
Child given grains in day/night before survey- last-born under 2 years	1.04	0.084
Child given roots or tubers in day/night before survey- last-born under 2 years	1.07	0.065
First marriage by age 22	1.18**	0.065
Child given eggs in day/night before survey- last-born under 2 years	1.18**	0.053
Number of children under 12 years old	1.27***	0.023
Started breastfeeding within one hour of birth - last-born in the past 2 years	1.27***	0.057
Child given other fruits or vegetables in day/night before survey- last-born under 2 years	0.72***	0.056
Attended ANC <4 months of pregnancy	0.71***	0.059
as.factor(year)		
2007	—	—
2008	0.87	0.171
2009	0.73*	0.156
2010	0.72*	0.154
2011	0.70*	0.156
2012	0.47***	0.164
2013	0.60**	0.159
2014	0.47***	0.158
2015	0.51***	0.145
2016	0.43***	0.146

[1] *p<0.05; **p<0.01; ***p<0.001

Abbreviations: CI = Confidence Interval, OR = Odds Ratio, SE = Standard Error